claude-windows / d4b5afb5-2f03-4b58-8952-573f1381d166

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\d4b5afb5-2f03-4b58-8952-573f1381d166\subagents\workflows\wf_beb40705-90b\agent-ac9b56c84593a55cc.jsonl`
- SHA-256: `98189cc35e2fb6e24b793614ff1f6399583fbfc908c4ca21caa5e5cf0e98e983`
- Source modified: `2026-06-21T03:19:07+00:00`
- Imported at: `2026-07-05T16:48:23+00:00`
- project: `wf_beb40705-90b`
- session_id: `d4b5afb5-2f03-4b58-8952-573f1381d166`

Transcript

1. user (2026-06-21T03:17:18.117Z)

```
You are ramping up an engineer on the ALREADY-WRITTEN i18n design for KhelSutra (a local AI
badminton highlight indexer, Bangalore-India market).
Read VERY carefully and completely: C:/Users/avidu/Projects/badminton-highlight-
indexer/docs/I18N_PLAN.md (229 lines).
Produce a thorough markdown brief capturing, with exact section/line refs where useful:
- The string-externalization model (t("key") usage, key naming convention, ICU MessageFormat
plurals).
- The catalog layout (en source catalog + per-locale siblings; file format/paths the doc
proposes).
- The locale-negotiation order (persisted pref -> switcher/?lang= -> Accept-Language -> en
fallback).
- The translation pipeline: Gemini-DRAFT -> human-REVIEW (note the guardrail: inference only,
NEVER training).
- CI enforcement: key-parity + no-bare-strings lint.
- Fonts: Noto Sans Kannada/Devanagari, the OFL licensing decision/open question.
- The phasing P0..P4 (esp. P3 = SPA react-i18next, P4 = conversational/coach-mode).
- ANY assumption the doc bakes in that is now STALE because it targeted the OLD static UI
(backend/api/static/*) rather than today's surfaces (a Vite/React/TS web/ SPA + a vanilla-HTML
site/). Flag each stale assumption explicitly.
- The 4 open OWNER decisions the doc raises in its section 4 (font licensing, v1 locale set,
backend-localization depth, UX scope/which-surface-first) ? quote them.
Return the brief as your final message (it is data for an orchestrator, not a human chat reply).
Be exhaustive and faithful; do not invent.
```

2. assistant (2026-06-21T03:17:20.473Z)

```
I'll read the i18n plan document carefully, and also check the actual current UI surfaces to
verify which assumptions are stale.
```

3. user (2026-06-21T03:17:21.376Z)

```
1       # Internationalization (i18n) ? Kannada-first, by design
2
3       **Status:** Design (P0 deliverable; phases P1+ are owner-gated, not started).
4       **Owner decision required** before any code lands ? see [Open decisions](#open-
decisions-for-the-owner).
5
6       ---
7
8       ## 1. Context ? why this, why now
9
10      The PMF beachhead is **Bangalore badminton academies**
([`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md)).
11      Many academy coaches, parents, and players are far more comfortable in **Kannada** than
```

English.
12   A product that only speaks English adds friction exactly at the point of first contact.
So the
13   product must speak **Kannada first**, **Hindi next**, and extend to other Indian
languages
14   (Tamil, Telugu, Marathi, ?) **without a refactor**.
15
16   The owner's directive is precise: make the product **internationalizable _by design_**.
Adding a
17   language must be *"add a reviewed catalog"* ? not *"go re-touch the code."* The
translations
18   themselves can wait; **the foundation that makes them cheap to add is the actual
deliverable.**
19
20   Today there is **zero i18n** in the repo (no i18n/locale/gettext/babel anywhere). Every
21   user-facing string is hardcoded English across four surfaces:
22
23 | Surface | Where | Examples (today) |
24 |---|---|---|
25 | **UI chrome** | `backend/api/static/{index.html,app.js,style.css}` (~40?50 strings) |
`"Validate & Index Video"`, `alert("Validation Failed! ?")`, ``` `${count} Rally Segment${count
> 1 ? 's' : ''} Selected` ``` |
26 | **Backend coach-facing text** | `backend/main.py` `HTTPException(detail=?)`;
`backend/pipeline/validators/*` `message`; `backend/reporting/spec.yaml` `customer_verdict` +
per-rule `customer_message` | `"Video file not found at '?'"`; `"Video quality rejected: Focus
check failed?"`; `"? Your highlights are ready."` |
27 | **Self-navigating NL queries** (owner ask, future) | not built yet | *"show me all
rallies > 15 shots where I won"* ? the **query and its answer** must work in the user's language
|
28 | **Coach-mode recommendations** (owner ask, future) | not built yet | localized
recommendation text, not English |
29
30   A React/Vite SPA is planned to replace the static UI ([`UI_ALPHA_EXECUTION_PLAN.md`],
ADR-009)
31   in a **separate private repo `avidullu/khelsutra`**. The design below is therefore
deliberately
32   **framework-agnostic**: the catalog format and locale-negotiation contract are shared,
so the
33   static UI today and the SPA tomorrow consume the *same* translation source.
34
35   **Scope confirmed with owner:** UI + coach-facing backend text + (designed-for, built-
later)
36   NL/coach features. **Translations are Gemini-drafted, human-reviewed** ? this is
inference
37   use of a paid LLM, which the [licensing guardrail](#guardrail-alignment) explicitly
allows
38   (it is *not* a training-data source). The owner reviews the first draft of every string.
39
40   ---
41
42   ## 2. Core architecture
43
44   ### 2.1 The "by design" invariant ? no hardcoded user-facing strings
45
46   Every coach-facing string is a **stable key** resolved at render time against the active
locale.
47   Code references `t("compile.button")`, never the English prose. This is the single rule
that
48   makes "add a language = add a catalog" true, and it is **enforced by CI** (§2.6) so it
cannot
49   silently regress as the codebase grows.
50
51   ```
52   "Validate & Index Video"   ? t("video.action.validate_index")
53   alert("Validation Failed! ?review the checklist") ? t("toast.validation_failed")

```
54      `${count} Rally Segment${count>1?'s':''} Selected`?  t("compile.selected", {count})   #
ICU plural
55      HTTPException(detail="Video file not found at ?")?  detail = tr(lang,
"error.video_not_found", {path})
56      ```
57
58      ### 2.2 One canonical English source catalog, per-locale siblings
59
60      ```
61      locales/
62        en/  common.json   # source of truth ? devs author keys here
63        kn/  common.json   # Kannada  (Gemini-draft ? human-review)
64        hi/  common.json   # Hindi    (next; first exercise of the extensibility checklist)
65      ```
66
67      - **Format: i18next-flavored JSON** ? works for the vanilla shim today *and*
react-i18next in
68        the SPA, with no conversion.
69      - **ICU MessageFormat** for interpolation and plurals: `"{count, plural, one {# rally}
other {# rallies}}"`.
70        This matters immediately ? the current UI already hand-rolls English plurals
71        (`Rally Segment${count > 1 ? 's' : ''}`), and Kannada/Hindi plural rules differ from
English.
72        ICU encodes each language's rules in the catalog, not the code.
73      - `en` is the source of truth; **every other locale is keyed identically** (enforced by
the
74        key-parity check in §2.6).
75
76      ### 2.3 Locale-negotiation contract (shared frontend ? backend)
77
78      One resolution order, defined **in one place** and honored by both UI and backend:
79
80      ```
81      persisted user preference  ?  UI switcher / ?lang=<code>  ?  Accept-Language header  ?
default "en"
82      ```
83
84      - Supported locales are enumerated in **a single module** (one list to grow). Anything
outside
85        it falls back to `en`.
86      - The backend localizes its own responses (`HTTPException.detail`, report `customer_*`)
from the
87        same `lang` query param / `Accept-Language` header, so a Kannada UI gets Kannada
errors.
88      - **Always-fallback to `en`** for any missing key ? a missing translation must degrade
to
89        English, never to a blank or a raw key in front of a user.
90
91      ### 2.4 Backend text ? keys + params, not prose
92
93      Validator `message`s and API errors today bake in English prose *and* interpolated
values
94      (`f"?sharpness score of {avg} is below ? threshold of {threshold}."`). The design
separates
95      the two:
96
97      - The validator/endpoint emits a **stable code + a params dict**
(`("blur.below_threshold",
98        {"score": avg, "threshold": threshold})`).
99      - The **prose lives in the catalog**, looked up at the API boundary in the negotiated
locale.
100     - `ValidationResult` keeps an English `message` for logs/back-compat, but gains an
optional
101       `message_key` + `message_params`; the report/serializer prefers the key when present.
This
```

```
102       keeps internal/technical messages English (good for debugging) while customer-facing
text
103       localizes.
104
105     `spec.yaml` `customer_verdict` and per-rule `customer_message` become **keyed** (or per-
locale
106     overlays keyed by rule `code`); the technical `message` blocks stay English (they cite
107     `README#Qn` and are for the operator, not the end user).
108
109     ### 2.5 Indic font strategy ? (open decision ? see §4)
110
111     The bundled UI/watermark fonts (Outfit, Roboto Slab) **do not cover Kannada or
Devanagari**.
112     **Noto Sans Kannada / Noto Sans Devanagari are OFL-licensed**, which is the same tension
we hit
113     with the watermark font: OFL is not in the strict **MIT/Apache/BSD** code/data/weights
guardrail.
114     Fonts are *media assets*, not code or model weights, and OFL is explicitly designed for
115     unrestricted redistribution ? so the cleanest path is to **carve out an explicit OFL-
for-fonts
116     exception**, the same way we'd treat any bundled media. Options put to the owner in §4.
117
118     Note the **video watermark already takes a `font_path`** (`WatermarkConfig.font_path`,
119     `backend/pipeline/stitcher/compiler.py`), so stamping a Kannada/Hindi watermark is a
120     *configuration* change (point it at a Noto font) once the font decision is made ? no new
code.
121
122     ### 2.6 CI enforcement (the guardrail that keeps it "by design")
123
124     Two checks, both pure-Python and runnable in this repo (no GPU/torch needed):
125
126     1. **Key-parity test** ? every locale catalog has *exactly* the `en` key set; fail on
missing
127        keys, extra keys, or untranslated placeholder markers (e.g. a `"__MISSING__"`
sentinel left
128        by the draft script). This is also the **extensibility proof**: drop a stub `ta.json`
with
129        one key and CI immediately lists every gap.
130     2. **No-bare-strings lint** ? a check that flags user-facing string literals not routed
through
131        `t()` / the backend `tr()` (scoped to the UI handlers and the API/validator/report
text
132        paths). Starts as a curated allowlist so it can land without boiling the ocean.
133
134     ### 2.7 Translation pipeline
135
136     ```
137     dev authors en keys  ?  scripts/i18n_draft.py  (Gemini, INFERENCE only)  ?  kn/hi draft
138                          ?  native-speaker review (owner reviews first draft)
139                          ?  CI key-parity + placeholder check  ?  merge
140     ```
141
142     `scripts/i18n_draft.py` reads `locales/en/*.json`, asks Gemini for each missing key in
the
143     target locale **with the key name and any ICU placeholders preserved**, and writes a
draft
144     catalog with every machine-drafted value flagged for review. It never touches `en`,
never
145     invents keys, and is a *drafting aid* ? a human reviews before merge. (Reuses the
existing
146     Gemini provider credentials/config; this is the same "paid LLM = inference, never
training"
147     posture as the rest of the system.)
148
149     ---
```

```
150
151     ## 3. Phased execution (owner-gated; ONE PR per slice, off `master`)
152
153     | Phase | Scope | Repo | Gate |
154     |---|---|---|---|
155     | **P0** | **This design doc** (`docs/I18N_PLAN.md`) | this | _the immediate
deliverable_ |
156     | **P1** | Foundation + **static-UI retrofit**: `locales/` + `en` source catalog
(extract the ~40?50 UI strings ? keys), a tiny vanilla **`t(key, vars)`** shim + `data-i18n`
wiring in `index.html`/`app.js`, **language switcher (en/kn)** + `localStorage` persistence +
`?lang=`, the font decision applied, and the **`kn`** catalog (Gemini-draft ? owner review).
Reuse the existing static mount / `serve_index` in `backend/main.py`. | this | owner approval |
157     | **P2** | **Backend + report localization**: `backend/i18n/` with `tr(lang, key,
params)` (same catalog format), refactor validator `message`s to keys+params, localize
`HTTPException.detail` + `spec.yaml` `customer_verdict`/`customer_message`, wire `Accept-
Language`/`?lang=`. | this | owner approval |
158     | **P3** | **SPA adoption**: **react-i18next** consuming the *same* `locales/` catalogs;
documented FE?BE contract. | `avidullu/khelsutra` | needs repo access; or owner/Opus executes
there |
159     | **P4** | **Conversational + coach-mode** (future): locale-aware NL query intake ?
**templated localized answers**; coach-mode recommendation **templates**. Ties to the owned-
model / PLATFORM §5A + rich-metadata work. | this + model | gated on those features existing |
160
161     **Extensibility checklist** (this *is* the by-design contract ? runs the same every
time):
162     > add `locales/<code>/common.json` ? `scripts/i18n_draft.py --locale <code>` ? human
review ?
163     > add `<code>` to the supported-locales list ? confirm font coverage ? CI key-parity
green.
164
165     **Hindi (`hi`) is the first real exercise of this checklist** ? if adding `hi` requires
touching
166     anything other than the steps above, the design failed and we fix the design, not paper
over it.
167
168     ---
169
170     ## 4. Open decisions (for the owner)
171
172     1. **Target-UI scope** ? confirm the *shared framework-agnostic foundation* (prove on
the static
173        UI now, SPA adopts the same catalogs later). Alternative: scope only to the SPA and
skip the
174        static-UI retrofit. _Recommendation: shared foundation ? it's the cheap, low-risk way
to
175        validate the catalog/negotiation contract before the SPA exists._
176     2. **Indic font licensing** ? (a) **allow OFL for bundled fonts** and self-host Noto
Sans
177        Kannada/Devanagari _(recommended; treats fonts as media, not code/weights)_; (b) rely
on
178        **system Indic fonts** (no bundle, inconsistent rendering across machines); (c)
**Google Fonts
179        CSS** (runtime external dependency, privacy/offline concerns for a local-first
product). This
180        also unblocks the **Kannada/Hindi video watermark** (needs a Noto `font_path`).
181     3. **v1 locale set & fallback** ? confirm `en` + `kn` for v1, `hi` next; always-fallback
to `en`
182        for missing keys.
183     4. **Backend-localization depth for v1** ? recommendation: localize the **customer
verdict /
184        report `customer_*`** and **validation rejection reasons** first (highest user
impact);
185        defer technical/operator messages (they stay English by design).
186
187     ---
```

```
188
189     ## 5. Key files / seams
190
191     - **Localize:** `backend/api/static/{index.html,app.js,style.css}`; `backend/main.py`
192       (`HTTPException.detail` + the `serve_index`/static mount);
`backend/pipeline/validators/*`
193       (`message` ? key+params; see `validators/base.py::ValidationResult`);
194       `backend/reporting/spec.yaml` (`customer_verdict` + per-rule `customer_message`).
195     - **New:** `locales/{en,kn,hi}/`, `backend/i18n/` (`tr()` + supported-locales +
negotiation),
196       `scripts/i18n_draft.py`, a CI key-parity test under `tests/` + a no-bare-strings check
wired
197       into `.github/workflows/ci.yml`.
198     - **Fonts:** reuse the bundled-asset pattern in `backend/assets/fonts/` (Apache-2.0
Roboto Slab
199       ships there today); add Noto Indic fonts there *iff* decision §4.2(a).
200     - **SPA:** `avidullu/khelsutra` (react-i18next) ? separate repo.
201
202     ---
203
204     ## 6. Verification
205
206     - **CI key-parity test** (Python, this repo): every locale has every `en` key; fail on
207       missing/extra/placeholder. Drop a stub `ta.json` ? CI flags every gap (the
extensibility proof).
208     - **Static UI:** load `?lang=kn` ? rendered strings resolve from the `kn` catalog;
switcher
209       persists across reload; any missing key falls back to English (no raw keys shown).
210     - **Backend:** request with `Accept-Language: kn` (or `?lang=kn`) ? localized error
`detail` +
211       report `customer_*`; unit tests for fallback, ICU interpolation, and plural rules.
212     - **Fonts:** Kannada + Devanagari render correctly ? **manual eyeball on the owner's
machine**
213       (the cloud container can't render fonts).
214     - **Docs-only P0 PR:** internal links resolve; indexed in [`docs/README.md`](README.md);
215       cross-linked from [`PLATFORM_ARCHITECTURE.md`](PLATFORM_ARCHITECTURE.md) (the NL/coach
surfaces)
216       and [`COMMERCIALIZATION.md`](COMMERCIALIZATION.md) (the Bangalore beachhead).
217
218     ---
219
220     ## Guardrail alignment
221
222     - **Paid LLM = inference only.** Gemini drafts translations (inference); it is **never**
a
223       training-data source. ? consistent with [`CLAUDE.md`](../CLAUDE.md) and the ratified
guardrail.
224     - **Licensing.** Code stays MIT/Apache-2.0/BSD. The *only* tension is **fonts** (OFL) ?
handled
225       as an explicit, owner-approved media-asset exception in §4.2, not a silent dependency.
226     - **No minors / biometrics scope creep.** i18n touches presentation only; it introduces
no new
227       personal-data collection.
228
229     [`UI_ALPHA_EXECUTION_PLAN.md`]: ./UI_ALPHA_EXECUTION_PLAN.md
230
```

4. user (2026-06-21T03:17:25.320Z)

```
Your configuration specifies to merge with the ref 'refs/heads/docs/doc-status-register'
from the remote, but no such ref was fetched.
---STATUS---
backend/api/static
STATIC EXISTS
NO web/
```

```
NO site/
```

5. assistant (2026-06-21T03:17:28.361Z)

Now let me verify the actual current state of the UI surfaces to ground the stale-assumption analysis. Let me check the static UI, the khelsutra SPA, and look for any `web/` or `site/` style surfaces.

6. user (2026-06-21T03:17:31.906Z)

```
=== static UI dir ===
total 48
drwxr-xr-x 1 avidu 197609     0 Jun 11 20:44 .
drwxr-xr-x 1 avidu 197609     0 Jun 19 13:03 ..
-rw-r--r-- 1 avidu 197609 16350 Jun 11 20:44 app.js
-rw-r--r-- 1 avidu 197609  8743 Jun  7 14:14 index.html
-rw-r--r-- 1 avidu 197609 13418 Jun  7 14:14 style.css
=== fonts dir ===
LICENSE-Apache-2.0.txt
README.md
RobotoSlab-Regular.ttf
=== locales? ===
NO locales/ yet
=== scripts i18n? ===
NO i18n_draft.py yet
=== backend i18n? ===
NO backend/i18n yet
```

7. user (2026-06-21T03:17:34.295Z)

```
=== khelsutra root ===
total 38
drwxr-xr-x 1 avidu 197609    0 Jun 20 23:27 .
drwxr-xr-x 1 avidu 197609    0 Jun 21 06:51 ..
drwxr-xr-x 1 avidu 197609    0 Jun 20 15:20 .claude
drwxr-xr-x 1 avidu 197609    0 Jun 21 08:47 .git
-rw-r--r-- 1 avidu 197609  297 Jun 20 23:27 .gitattributes
drwxr-xr-x 1 avidu 197609    0 Jun 20 12:43 .github
-rw-r--r-- 1 avidu 197609   78 Jun 20 18:40 .gitignore
-rw-r--r-- 1 avidu 197609 5823 Jun 20 23:06 NEXT_STEPS.md
-rw-r--r-- 1 avidu 197609  925 Jun 20 08:37 README.md
drwxr-xr-x 1 avidu 197609    0 Jun 20 23:14 docs
drwxr-xr-x 1 avidu 197609    0 Jun 21 08:20 site
drwxr-xr-x 1 avidu 197609    0 Jun 20 22:08 web
=== web/ ? ===
web
README.md
dist
index.html
node_modules
package-lock.json
package.json
src
tsconfig.json
vite.config.ts
=== site/ ? ===
site
README.md
access.db
make_og.py
node_modules
package-lock.json
package.json
public
```

```
schema.sql
scripts
server.js
=== vite/package ===
=== git remote ===
origin  https://github.com/avidullu/khelsutra.git (fetch)
origin  https://github.com/avidullu/khelsutra.git (push)
```

8. assistant (2026-06-21T03:17:38.285Z)

```
The khelsutra repo confirms today's surfaces: a `web/` Vite/React/TS SPA and a separate `site/`
(Node `server.js` + `public/`). Let me inspect both to ground the stale-assumption flags
precisely.
```

9. user (2026-06-21T03:17:40.940Z)

```
=== web/package.json ===
{
  "name": "khelsutra-web",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  "description": "KhelSutra Studio ? operator console SPA (drives the rally indexer engine).",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "preview": "vite preview",
    "typecheck": "tsc --noEmit",
    "test": "vitest run"
  },
  "dependencies": {
    "react": "^19.0.0",
    "react-dom": "^19.0.0"
  },
  "devDependencies": {
    "@tailwindcss/vite": "^4.0.0",
    "@types/react": "^19.0.0",
    "@types/react-dom": "^19.0.0",
    "@vitejs/plugin-react": "^4.3.4",
    "tailwindcss": "^4.0.0",
    "typescript": "^5.7.0",
    "vite": "^6.0.0",
    "vitest": "^3.0.0"
  }
}
=== web/src tree ===
web/src:
App.tsx
api
components
hooks
index.css
lib
main.tsx

web/src/api:
client.test.ts
client.ts
types.ts

web/src/components:
CompileResult.tsx
QueryBar.examples.test.ts
QueryBar.tsx
RallyGrid.tsx
```

```
SelectionFooter.tsx

web/src/hooks:
useRallySelection.ts

web/src/lib:
floor.test.ts
floor.ts
format.test.ts
format.ts
log.ts
query.test.ts
query.ts
selection.test.ts
selection.ts
=== web has i18next? ===
no i18next in web/package.json
```

10. user (2026-06-21T03:17:42.567Z)

```
=== site/package.json ===
{
  "name": "khelsutra-site",
  "private": true,
  "type": "module",
  "scripts": {
    "test": "node --test",
    "test:cf": "node --test src/index.test.js",
    "test:vm": "node --test src/server.test.js",
    "test:deploy": "node --test src/deploy-config.test.js src/assets.test.js",
    "deploy": "wrangler deploy",
    "dev": "wrangler dev",
    "og": "python make_og.py"
  },
  "devDependencies": {
    "jsdom": "^29.1.1"
  }
}
=== site/public ===
capture.html
img
index.html
og.png
own-model.html
video
=== site type ===
// Always-on VM host for the KhelSutra site ? dependency-free (Node 22+ only).
// Serves public/ and POST /api/request-access into SQLite (node:sqlite), using the
// SAME handler logic as the Cloudflare Worker (src/index.js). This is the portability
// fallback "if Cloudflare acts out" ? see docs/PORTABILITY.md.
//
//   PORT=8080 DB_PATH=/var/lib/khelsutra/access.db node server.js
//
import { createServer } from "node:http";
import { readFile } from "node:fs/promises";
import { join, extname, normalize } from "node:path";
import { DatabaseSync } from "node:sqlite";
import { handleRequest } from "./src/index.js";
import { sqliteStore } from "./src/store.js";
import { noopNotify } from "./src/notify.js";

const PORT = Number(process.env.PORT) || 8080;
const ROOT = import.meta.dirname;
const PUBLIC = join(ROOT, "public");
const DB_PATH = process.env.DB_PATH || join(ROOT, "access.db");
```

```
const db = new DatabaseSync(DB_PATH);
db.exec(await readFile(join(ROOT, "schema.sql"), "utf8")); // CREATE TABLE IF NOT EXISTS ?
idempotent
const store = sqliteStore(db);
const notify = noopNotify(); // VM host: no email yet (add SMTP later ? see docs/PORTABILITY.md)

const TYPES = {
  ".html": "text/html; charset=utf-8", ".png": "image/png", ".svg": "image/svg+xml",
  ".css": "text/css", ".js": "text/javascript", ".ico": "image/x-icon", ".txt": "text/plain",
  ".mp4": "video/mp4", ".webm": "video/webm", ".gif": "image/gif", ".jpg": "image/jpeg",
".jpeg": "image/jpeg",
};
=== khelsutra README ===
```

KhelSutra Guru ?

```
**Private product repo** ? the heart and brain of [khelsutra.guru](https://khelsutra.guru):
the web app (alpha SPA) now, the hosted control plane at beta.

- **Engine** (pipeline, API, eval): sibling repo `badminton-highlight-indexer` (rename pending).
- **Canonical planning docs** (live in the engine repo's `docs/` for now, migrate here at beta):
  `UI_PROPOSAL.md` · `HOSTING_PLAN.md` · `UI_ALPHA_EXECUTION_PLAN.md`.
- **Status:** pre-alpha scaffold. SPA lands in `web/` per execution-plan milestone M1.
```

Structure

```
- `site/` ? public marketing landing page (static; served at khelsutra.guru)
- `web/` ? React SPA (alpha; Vite + React + TS + Tailwind; Cloudflare Pages build root)
- `docs/` ? product docs as they migrate from the engine repo
- `control-plane/` ? (beta) Cloud Run FastAPI control plane
```

License

```
All rights reserved (private; no OSS license).
```

11. assistant (2026-06-21T03:19:07.715Z)

```
I now have everything needed to ground the analysis. The doc's central assumption ? that the SPA
lives in a "planned" separate repo and that the static `backend/api/static/*` UI is "today's"
surface ? is confirmed stale: the `avidullu/khelsutra` repo already exists with a live `web/`
Vite/React/TS SPA and a separate `site/` vanilla-HTML marketing landing page. Here is the brief.

---
```

i18n Design Ramp-Up Brief ? KhelSutra (`docs/I18N_PLAN.md`, 229 lines)

```
**Doc status (line 3):** "Design (P0 deliverable; phases P1+ are owner-gated, not started)."
Owner sign-off required before any code lands (line 4). Tagline: **"Internationalization (i18n)
? Kannada-first, by design"** (line 1).

**One-line thesis (lines 16?18):** make the product internationalizable *by design* so that
adding a language is *"add a reviewed catalog"*, never *"go re-touch the code."* The
translations can wait ? **the foundation that makes them cheap to add is the actual P0
deliverable.**

**Market driver (lines 10?14):** PMF beachhead = Bangalore badminton academies;
coaches/parents/players often more comfortable in **Kannada**. Order: **Kannada first, Hindi
next**, then Tamil/Telugu/Marathi "without a refactor."

---
```

1. String-externalization model (§2.1, lines 44?56)

- **The invariant:** *no hardcoded user-facing strings.* Every coach-facing string is a **stable key** resolved at render time against the active locale. Code references `t("compile.button")`, **never the English prose** (lines 46?47). This single rule is what makes "add a language = add a catalog" true, and it is **CI-enforced** (§2.6) so it cannot silently regress.
- **`t("key")` frontend / `tr(lang, key, params)` backend** are the two accessors.
- **Key naming convention** ? dotted, hierarchical, namespaced-by-surface/intent. Exact examples the doc commits to (lines 52?56):
  - `"Validate & Index Video"` ? `t("video.action.validate_index")`
  - `alert("Validation Failed! ?")` ? `t("toast.validation_failed")`
  - `` `${count} Rally Segment${count>1?'s':''} Selected` `` ? `t("compile.selected", {count})` *(ICU plural)*
  - `HTTPException(detail="Video file not found at ?")` ? `detail = tr(lang, "error.video_not_found", {path})`
- **ICU MessageFormat for interpolation + plurals** (§2.2, lines 69?72): `"{count, plural, one {# rally} other {# rallies}}"`. Rationale called out explicitly: the current UI hand-rolls English plurals (`Rally Segment${count > 1 ? 's' : ''}`), and **Kannada/Hindi plural rules differ from English** ? ICU encodes each language's rules **in the catalog, not the code.**

2. Catalog layout (§2.2, lines 58?74)

```
locales/
  en/  common.json   # source of truth ? devs author keys here
  kn/  common.json   # Kannada  (Gemini-draft ? human-review)
  hi/  common.json   # Hindi    (next; first exercise of the extensibility checklist)
```

- **Format: i18next-flavored JSON** (line 67) ? deliberately chosen to work for **both** the vanilla shim today *and* react-i18next in the SPA, **with no conversion**.
- **`en` is the single source of truth**; every other locale is **keyed identically** to `en`, enforced by the key-parity check (§2.6) (lines 73?74).
- Proposed file/seam paths (§5, lines 195?199): new dirs `locales/{en,kn,hi}/`, `backend/i18n/`, `scripts/i18n_draft.py`; fonts under `backend/assets/fonts/`.

3. Locale-negotiation order (§2.3, lines 76?89)

One resolution order, defined in one place, honored by both UI and backend (line 81):

```
persisted user preference  ?  UI switcher / ?lang=<code>  ?  Accept-Language header  ?  default "en"
```

- Supported locales enumerated in **a single module** (one list to grow); anything outside it falls back to `en` (lines 84?85).
- Backend localizes its **own** responses (`HTTPException.detail`, report `customer_*`) from the *same* `lang` query param / `Accept-Language` header, so a Kannada UI gets Kannada errors (lines 86?87).
- **Always-fallback to `en`** for any missing key ? a missing translation degrades to English, never to a blank or a raw key shown to a user (lines 88?89).

4. Backend text ? keys + params (§2.4, lines 91?107)

- Today validators/errors bake English prose *and* interpolated values together (`f"?sharpness score of {avg} is below ? threshold of {threshold}."`). Design **separates the two**:
  - Validator/endpoint emits a **stable code + params dict** ? e.g. `("blur.below_threshold", {"score": avg, "threshold": threshold})` (lines 97?98).
  - **Prose lives in the catalog**, looked up at the API boundary in the negotiated locale (line 99).
  - `ValidationResult` keeps an **English `message`** for logs/back-compat but **gains optional `message_key` + `message_params`**; the report/serializer prefers the key when present. Internal/technical messages stay English (debugging) while customer-facing text localizes (lines 100?103).
- `spec.yaml` `customer_verdict` + per-rule `customer_message` become **keyed** (or per-locale

overlays keyed by rule `code`); the technical `message` blocks stay English (they cite `README#Qn`, are for the operator) (lines 105?107).

5. Translation pipeline ? Gemini-DRAFT ? human-REVIEW (§2.7, lines 134?147; also §1 lines 35?38)

```
dev authors en keys ? scripts/i18n_draft.py (Gemini, INFERENCE only) ? kn/hi draft
                    ? native-speaker review (owner reviews first draft)
                    ? CI key-parity + placeholder check ? merge
```

- `scripts/i18n_draft.py` reads `locales/en/*.json`, asks Gemini for each **missing** key in the target locale **with the key name and any ICU placeholders preserved**, and writes a draft catalog with **every machine-drafted value flagged for review** (lines 142?144).
- It **never touches `en`, never invents keys**, and is a *drafting aid only* ? a human reviews before merge (lines 144?145).
- **GUARDRAIL (load-bearing, stated twice ? lines 37?38, 146?147, and §"Guardrail alignment" lines 222?223):** Gemini is used for **inference only** (drafting); it is **NEVER a training-data source.** This is the same "paid LLM = inference, never training" posture as the rest of the system, consistent with `CLAUDE.md` and the ratified licensing guardrail. The owner reviews the first draft of **every** string.

6. CI enforcement (§2.6, lines 122?132) ? "the guardrail that keeps it by design"

Two checks, **both pure-Python, runnable in this repo (no GPU/torch)** (line 124):

1. **Key-parity test** (lines 126?129) ? every locale catalog has *exactly* the `en` key set; **fail on missing keys, extra keys, or untranslated placeholder markers** (e.g. a `"__MISSING__"` sentinel left by the draft script). Doubles as the **extensibility proof**: drop a stub `ta.json` with one key ? CI immediately lists every gap.
2. **No-bare-strings lint** (lines 130?132) ? flags user-facing string literals **not routed through `t()` / backend `tr()`** (scoped to UI handlers + API/validator/report text paths). **Starts as a curated allowlist** so it can land without "boiling the ocean."

Wired into `.github/workflows/ci.yml`; tests under `tests/` (§5, lines 196?197). Verification restates the parity test + the stub-`ta.json` extensibility proof (§6, lines 206?207).

7. Fonts ? Indic font strategy (§2.5, lines 109?120; decision in §4.2)

- Bundled UI/watermark fonts (**Outfit, Roboto Slab**) **do not cover Kannada or Devanagari** (line 111).
- **Noto Sans Kannada / Noto Sans Devanagari are OFL-licensed** (line 112) ? same tension as the watermark font: **OFL is not in the strict MIT/Apache/BSD** code/data/weights guardrail.
- **Recommended resolution:** fonts are *media assets*, not code or model weights; OFL is designed for unrestricted redistribution ? **carve out an explicit OFL-for-fonts exception**, treated like any bundled media (lines 114?116).
- **Watermark is already config-only** (lines 118?120): the video watermark already takes a `font_path` (`WatermarkConfig.font_path`, `backend/pipeline/stitcher/compiler.py`), so a Kannada/Hindi watermark is just pointing it at a Noto font once the decision is made ? **no new code**.
- Today's bundled-font asset pattern lives at `backend/assets/fonts/` (currently Apache-2.0 Roboto Slab); add Noto Indic there **iff** decision §4.2(a) (§5, lines 198?199). *(Verified on disk: `backend/assets/fonts/` contains only `RobotoSlab-Regular.ttf` + `LICENSE-Apache-2.0.txt` ? no Noto fonts yet.)*

8. Phasing P0?P4 (§3, lines 151?166)

Phase	Scope	Repo	Gate
P0	**This design doc** (`docs/I18N_PLAN.md`)	this	the immediate deliverable
P1	Foundation + **static-UI retrofit**: `locales/` + `en` source catalog (extract ~40?50 UI strings ? keys); a **tiny vanilla `t(key, vars)` shim** + `data-i18n` wiring in `index.html`/`app.js`; **language switcher (en/kn)** + `localStorage` persistence + `?lang=`;

font decision applied; the **`kn`** catalog (Gemini-draft ? owner review). Reuse existing static mount / `serve_index` in `backend/main.py`. | this | owner approval |
P2	**Backend + report localization**: `backend/i18n/` with `tr(lang, key, params)` (same catalog format); refactor validator `message`s ? keys+params; localize `HTTPException.detail` + `spec.yaml` `customer_verdict`/`customer_message`; wire `Accept-Language`/`?lang=`.	this	owner approval
P3	**SPA adoption: react-i18next** consuming the *same* `locales/` catalogs; documented FE?BE contract.	`avidullu/khelsutra`	needs repo access; or owner/Opus executes there
P4	**Conversational + coach-mode** (future): locale-aware NL query intake ? **templated localized answers**; coach-mode recommendation **templates**. Ties to owned-model / PLATFORM §5A + rich-metadata work.	this + model	gated on those features existing

Extensibility checklist (lines 161?163) ? *this is the by-design contract*:
> add `locales/<code>/common.json` ? `scripts/i18n_draft.py --locale <code>` ? human review ? add `<code>` to supported-locales list ? confirm font coverage ? CI key-parity green.

Hindi (`hi`) is the first real exercise (lines 165?166): if adding `hi` requires touching anything beyond those steps, "the design failed and we fix the design, not paper over it."

---

9. STALE ASSUMPTIONS ? doc targets the OLD static UI, not today's surfaces

The doc was written when the only concrete UI was `backend/api/static/*` and the SPA was merely *planned in a "separate private repo."* **That repo now exists and is materially further along than the doc assumes.** Verified on disk in `C:\Users\avidu\Projects\khelsutra`: a live **`web/` Vite + React 19 + TS + Tailwind 4 SPA** (`web/package.json`, `web/src/App.tsx`, components, vitest) **and** a separate **`site/` vanilla-HTML marketing landing page** (Node `server.js` + `public/{index.html,capture.html,own-model.html,...}`, Cloudflare Worker `src/index.js`). Each stale assumption, flagged:

1. **"Today there is a static UI at `backend/api/static/*`; the SPA is *planned* / *tomorrow*" ? STALE.** Lines 20?33 frame `backend/api/static/{index.html,app.js,style.css}` as "the static UI today" and the SPA as "planned to replace the static UI ? in a separate private repo `avidullu/khelsutra`." Reality: the SPA already lives and runs in `khelsutra/web/`. The static UI is legacy/parallel, not "the present surface." Any P1 plan to *retrofit `backend/api/static/*` first* may be retrofitting a surface that is already being superseded.

2. **The "four surfaces" table omits the `site/` vanilla-HTML marketing page entirely ? STALE/INCOMPLETE.** §1 table (lines 23?28) enumerates UI chrome, backend coach-facing text, NL queries, coach-mode. It has **no row** for the public landing page. Today `khelsutra/site/public/` (`index.html`, `capture.html`, `own-model.html`, request-access form copy) is a real, user-facing, currently-shipping English surface ? and arguably the **first** thing a Bangalore academy sees. Its localization is unscoped by this doc.

3. **P1's "static-UI retrofit" target is misdirected ? STALE.** Line 156 spends P1 on extracting the ~40?50 strings from `backend/api/static/*` + a "tiny vanilla `t(key,vars)` shim" + `data-i18n` wiring in `index.html`/`app.js`, reusing `serve_index` in `backend/main.py`. With the SPA already built in React, the **vanilla-shim retrofit of the legacy static UI may be throwaway work.** The two genuinely-current frontend surfaces are (a) the React `web/` SPA (which needs react-i18next, i.e. doc's **P3**, not P1) and (b) the `site/` HTML (which needs *some* shim, but it's a different codebase/repo than the one P1 names).

4. **Phase ordering assumes static-UI (P1) before SPA (P3) ? STALE / likely inverted.** The doc gates P3 behind "needs repo access; or owner/Opus executes there" (line 158) as if the SPA is a future, access-blocked concern. Since `web/` exists now and has **no i18next** in its deps (verified: `web/package.json` lists react/vite/tailwind/vitest, *no* `i18next`/`react-i18next`), the real, current frontend i18n work is **P3-shaped and immediate**, not P1. The doc's "prove on the static UI first, SPA adopts later" sequencing (§4.1) predates the SPA existing.

5. **"Reuse the existing static mount / `serve_index` in `backend/main.py`" ? STALE coupling assumption.** Line 156. The current SPA is a Vite build (Cloudflare Pages build root per khelsutra README) and the landing page is served by Cloudflare Worker / Node `server.js` ? **not** FastAPI's `serve_index`. The frontend?backend serving topology the doc assumes (FastAPI

serves the HTML) no longer matches how `web/` and `site/` are actually served.

6. **§5 "Key files / seams" and §1 examples point only at `backend/api/static/*` ? STALE inventory.** Lines 191?200 list `backend/api/static/{index.html,app.js,style.css}` as *the* frontend localize-targets. The real frontend string inventory now lives in `khelsutra/web/src/components/*` (`QueryBar.tsx`, `RallyGrid.tsx`, `CompileResult.tsx`, `SelectionFooter.tsx`, `App.tsx`) and `khelsutra/site/public/*.html`. The string-extraction estimate ("~40?50 UI strings") is measured against the obsolete static UI, not the SPA + landing page.

7. **Repo-of-record framing is dated ? note.** Lines 30?33/200 call `avidullu/khelsutra` a "separate private repo" the SPA is "planned to" move to. It already *is* the SPA's home and is checked out locally as a sibling (`C:\Users\avidu\Projects\khelsutra`, remote `github.com/avidullu/khelsutra`). The "or owner/Opus executes there" hedge (line 158) is no longer access-blocked in practice.

> **What is NOT stale (still valid, and the doc's actual durable value):** the framework-agnostic *catalog format* (i18next JSON), the *locale-negotiation contract*, ICU plurals, the Gemini-draft?human-review pipeline with the inference-only guardrail, the CI key-parity + no-bare-strings checks, the backend keys+params refactor (§2.4 ? `backend/` is the same repo and unchanged), and the font decision. These are explicitly designed to be shared across "the static UI today and the SPA tomorrow" (lines 31?33), so they survive the surface change. The breakage is purely in **which frontend surface P1 targets and the P1-before-P3 ordering**, because the SPA arrived ahead of the doc.

---

10. The 4 open OWNER decisions (§4, lines 170?185) ? quoted verbatim

> **1. Target-UI scope** ? confirm the *shared framework-agnostic foundation* (prove on the static UI now, SPA adopts the same catalogs later). Alternative: scope only to the SPA and skip the static-UI retrofit. _Recommendation: shared foundation ? it's the cheap, low-risk way to validate the catalog/negotiation contract before the SPA exists._

> **2. Indic font licensing** ? (a) **allow OFL for bundled fonts** and self-host Noto Sans Kannada/Devanagari _(recommended; treats fonts as media, not code/weights)_; (b) rely on **system Indic fonts** (no bundle, inconsistent rendering across machines); (c) **Google Fonts CSS** (runtime external dependency, privacy/offline concerns for a local-first product). This also unblocks the **Kannada/Hindi video watermark** (needs a Noto `font_path`).

> **3. v1 locale set & fallback** ? confirm `en` + `kn` for v1, `hi` next; always-fallback to `en` for missing keys.

> **4. Backend-localization depth for v1** ? recommendation: localize the **customer verdict / report `customer_*`** and **validation rejection reasons** first (highest user impact); defer technical/operator messages (they stay English by design).

Note for the owner-decision discussion: Decision #1 is the one most directly invalidated by §9 above ? its premise ("prove on the static UI now ? before the SPA exists") is counterfactual: the SPA exists. The decision should be re-framed as *which of the two live surfaces (`web/` SPA via react-i18next, `site/` landing via a shim) leads*, and whether the legacy `backend/api/static/*` retrofit is worth doing at all.

Source references

- Primary doc (read in full): `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\I18N_PLAN.md` (229 lines).
- Stale-assumption grounding (verified on disk): `C:\Users\avidu\Projects\khelsutra\web\` (Vite+React 19+TS+Tailwind SPA, `web/package.json`, `web/src/components/*`; **no i18next dependency**), `C:\Users\avidu\Projects\khelsutra\site\` (vanilla-HTML landing, `site/server.js` + `site/public/*.html`), `C:\Users\avidu\Projects\khelsutra\README.md`.
- Legacy static UI still present: `C:\Users\avidu\Projects\badminton-highlight-indexer\backend\api\static\{index.html,app.js,style.css}`.

- Fonts seam: `C:\Users\avidu\Projects\badminton-highlight-indexer\backend\assets\fonts\` (only `RobotoSlab-Regular.ttf` + Apache-2.0 license today; **no Noto Indic fonts yet**).
- Not-yet-created scaffolding the doc proposes: `locales/`, `backend/i18n/`, `scripts/i18n_draft.py` ? **none exist yet** (P1+ not started, consistent with the doc's "owner-gated, not started" status).